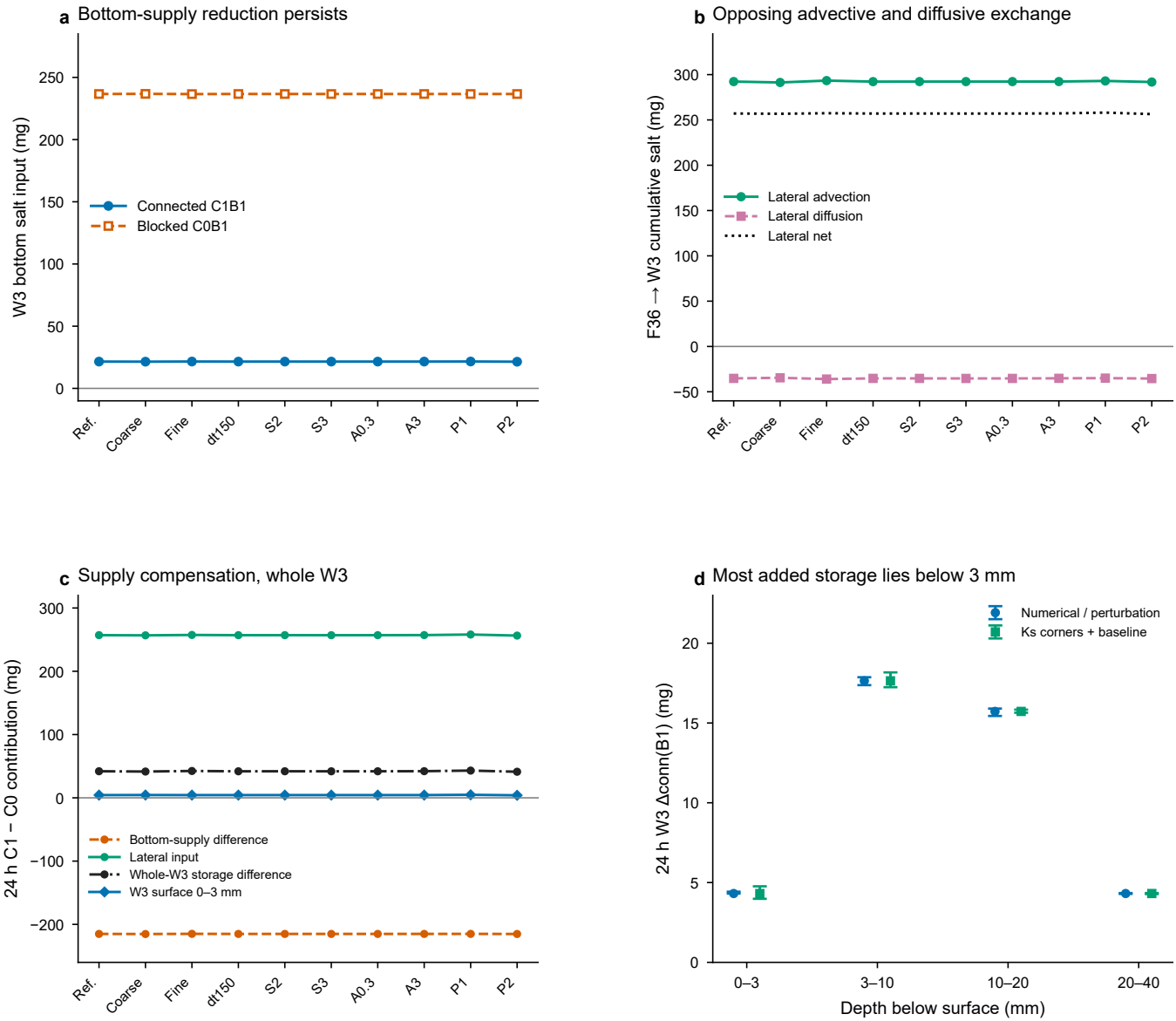

B1 only; all panels use the same 10 scenarios. S2/S3: phases 2/3; A0.3/A3: amplitude ×0.3/×3; P1/P2: conductivity corners.
Lines across scenario categories guide comparison; categories are not a continuous parameter sweep. Depth whiskers show min–max ranges, not CIs.
Whole-side budgets do not substitute for a surface-layer budget. Surface budgets are shown separately in Fig. S local transport.